Centro de Investigación en Complejidad Social · Universidad del Desarrollo · Chile

Networks Decide for Us

Imputed homophilic networks and the structural premium of diffusion

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The question

- Collective behaviors — innovations, mobilizations — spread as **contagions** on social networks.
- Real human networks have a specific texture: we cluster with people like us — **multidimensional homophily** [1].
- Yet most diffusion models run on *synthetic* topologies, and most surveys give us *individuals* but *not their networks*.

Holding individual propensities *fixed*, does the **shape of the network alone** — not who its members are — decide *how much* collective behavior spreads?

The gap, and where we go

The standing problems

- Diffusion ABMs use *made-up* networks (small-world, scale-free) [2].
- Rich surveys (ATP, GSS) measure *propensities* but have *no* relational data.
- Threshold models often assume *all* adoption needs peer influence [3].

Our two moves

- **Impute** plausible whole-networks onto real survey respondents from empirical homophily.
- **Measure** the causal premium of that structure on diffusion, and ask *what about it* matters.

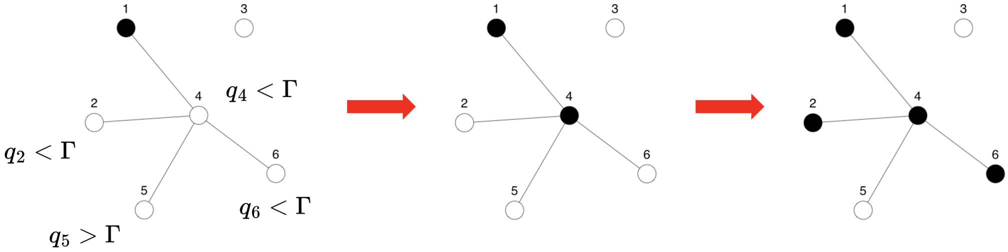
Building on *imputing social context from survey data* [4] — but going from a latent context term to **full simulable networks**.

The model

Adoption, part 1 — Rational Choice

- A behavior has an *Intrinsic Utility Level* (IUL) $\Gamma \in [0,1]$ — a latent quality scale: the highest-IUL option is the one that wins the most **pairwise comparisons** [5].
- Each person has a *Minimum Utility Requirement* $q_i \in [0,1]$ (their aversion).

An agent adopts on its own iff $\boxed{q_i \leq \Gamma \implies a_i = 1}$



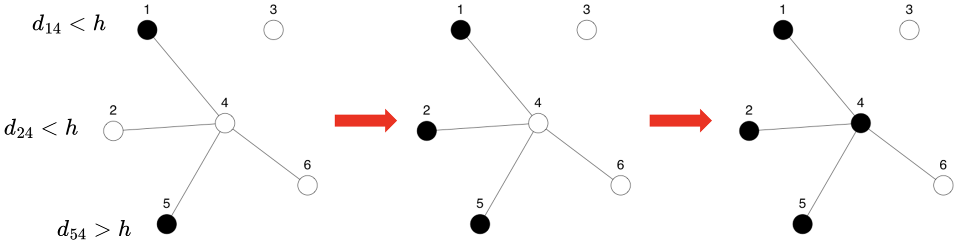
This channel is **topology-blind** — it does not need the network at all.

Adoption, part 2 — Selective Social Influence

Otherwise, an agent adopts by peer contagion if $\tau_i \leq \tilde{E}_i$ [3] — but only peers who are socially *close* [6] count:

$$\tilde{E}_i \equiv \frac{\sum_{j \neq i} \mathbf{X}_{ij}\, \tilde{a}_j}{\sum_{j \neq i} \mathbf{X}_{ij}}, \qquad \tilde{a}_j = 1 \iff a_j = 1 \wedge U_{ij} \leq \sigma(MSP - d_{ij})$$

- d_{ij}: social (Blau-space) distance; **MSP** (Maximum Social Proximity) sets how far across demographic lines influence can reach.



Imputing networks onto a survey

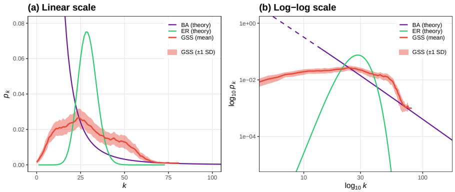
The imputation pipeline

Empirical homophily strengths $\hat{\boldsymbol{\beta}}$ [6,4] → Simulate whole networks via **ERGM** → Attach survey propensities q_i to each node

Tie probability is a function of distance in Blau space (age, sex, education, race, religion):

$$\text{logit } \Pr(Y_{ij} = 1) = \alpha + \boldsymbol{\beta}^\top \mathbf{d}_{ij}$$

$\hat{\boldsymbol{\beta}}$ transported from GSS confidant data [6]; propensities from ATP (innovation) and GSS (collective action).



Imputed networks reproduce the right-skewed degree structure of real networks — synthetic models don't.

Simulation design

Hybrid social contagion via `netdiffuseR` [7], over plausible (GSS) vs. **degree-preserving randomized** (GSS-DP) topologies.

- **IUL** (Γ): 41 levels. **MSP** (h): 13 levels.
- **Thresholds** $\tau_i \sim \mathcal{N}(\mu, \sigma)$; **seeding**: random, central, marginal, closeness, eigenvector.

$$41 \times 13 \times 4 \times 4 \times 5 \times \text{(runs)} \approx 4 \times 10^6 \textbf{ simulations}$$

The DP baseline keeps each person's *number* of ties but rewires *whom* they connect to — our counterfactual for "the same people, different structure."

Results

Result 1 — A structural premium

A Big Additive Model isolates the marginal effect of destroying empirical structure:

$$\boxed{\beta_{\text{DP}} = -0.850^{***} \quad (\text{OR} = 0.43)}$$

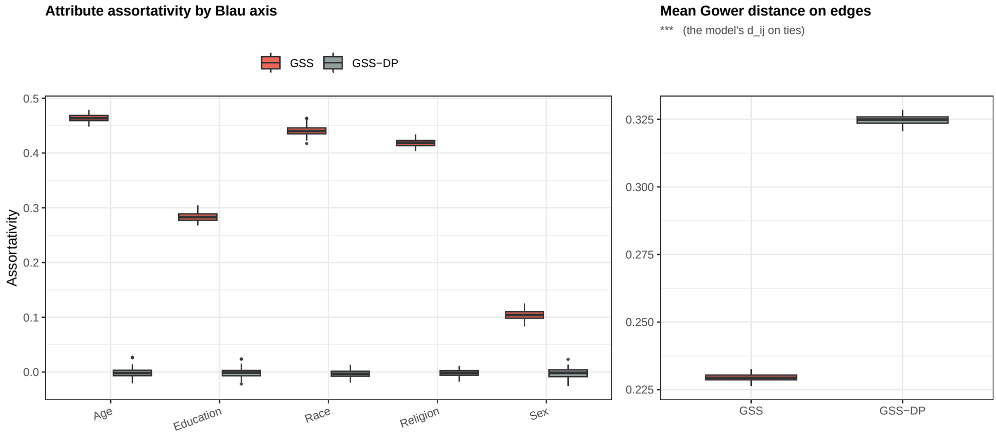
$\rightarrow$ randomization cuts adoption odds by $\sim$**57%**

Same people, same propensities — only the **wiring** changed. Homophilic clustering lets social reinforcement build inside local echo chambers.

Total adoption (GSS)	Estimate	OR
(Intercept)	6.534^{***}	—
τ_{mean}	-6.944^{***}	—
τ_{sd}	5.942^{***}	—
Seeding (ref: Random)		
Central	0.120^{***}	1.127
Closeness	0.119^{***}	1.126
Eigenvector	0.118^{***}	1.125
Marginal	-0.067^{***}	0.935
Topology (ref: Plausible)		
Deg.-pres. random	$\mathbf{-0.850}^{***}$	**0.427**
Deviance explained: 82.2%	$N = 4{,}093{,}440$	

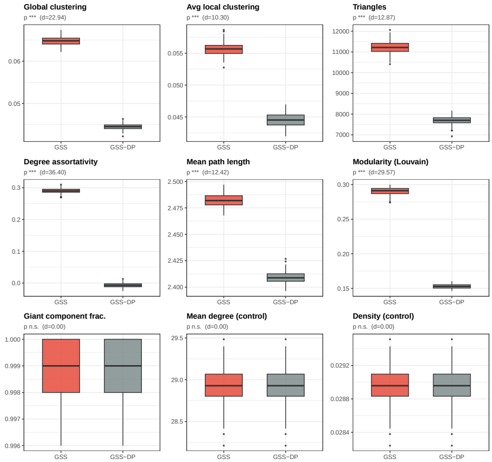
$^{***}p < 0.001$. OR $= \exp(\beta)$.

Result 2 — What does “breaking structure” destroy?



100 plausible vs. 100 DP networks. **Homophily is annihilated**: every Blau-axis assortativity collapses to ≈ 0, and the model’s own tie distance (d_{ij}, Gower) rises **+42%** — neighbors go from *similar* to *random*. All $p < 10^{-30}$.

Result 2 — Topology degrades too, but less completely



Clustering -31%, triangles -31%, modularity -47% — yet path length barely moves; degree, density, giant component identical (controls). **Homophily** is the prime suspect; isolating it from clustering is ongoing work (attribute-shuffle null).

Where this sits

Macy & Evtushenko (2020) [8]

- Reintroduce individual idiosyncrasy as *stochastic noise*.
- Noise can *spontaneously instigate* cascades a weak-interest group would otherwise doom.

Here

- Idiosyncrasy is **measured preference**, not noise.
- At low IUL, *rational* adopters are endogenous instigators — we go one level *behind* spontaneous instigation.
- And **structure**, not just the threshold distribution, governs the outcome.

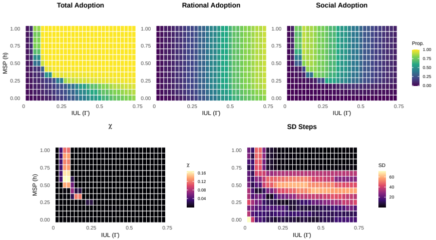
The same agents, on a different topology, produce a different society.

Teaser — the *shape* of change also differs

Beyond *how much*, structure governs *how* adoption tips:

- Plausible networks → a **continuous, predictable** transition.
- Randomized networks → an **abrupt, all-or-nothing** regime shift.

There is even a signature of **universality** (mean-field scaling) — a story for another talk [9]. Happy to discuss in Q&A.



Adoption & its variance over the IUL–MSP plane.

Conclusions

Networks decide for us.

1. **A method:** impute **simulable whole-networks** onto survey respondents from empirical homophily — reusable wherever you have propensities but no ties.
2. **A finding:** holding propensities fixed, empirical structure yields a $+57\%$ **adoption premium**.
3. **A mechanism:** what “breaking structure” destroys is, above all, **homophily** — neighbors stop being similar.

Macroscopic diffusion is not the sum of individual preferences — it is governed by the structure of our ties.

Limitations & next steps

- **Separate homophily from clustering:** the attribute-shuffle null model — identical topology, permuted node attributes — isolates pure homophily.
- **Heterogeneous MSP:** social porosity that varies by group, not a global constant.
- **Higher-stakes behavior:** extend the propensity scores to risky / high-cost collective action.
- **The criticality story:** the continuous-vs-abrupt transition and its universality (the teaser) — a companion paper.

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Thanks!

Scan: preprint draft (Zenodo)

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